

Technical Signal Validity and Structural Deterioration Framework

Institutional Version 1 assessment of RSI and Bollinger Band contribution on four-hour SOL/USDT data.

Executive determination. Under the frozen Version 1 specification, RSI, Bollinger Bands, and the combined specification did not demonstrate sufficiently stable incremental value to pass the establishment gate. The appropriate operational status for all three candidates is **NOT_ESTABLISHED**.

Assessment scope

Component	Version 1 specification
Research case	SOL/USDT spot; BTC/USDT market context; Binance; four-hour observations.
Candidate models	RSI, Bollinger Bands, and a secondary combined specification.
Validation	Five expanding chronological folds with a forecast-horizon gap and fold-local estimation.
Economic evidence	Incremental net edge after a 10 basis-point one-way cost assumption, with dependence-aware intervals.
Market states	Training-only filtered range, trend, and stress probabilities.
Monitoring	Robust one-sided CUSUM applied only after historical establishment.

Evidence summary

Candidate	Positive folds	Mean log-loss contribution	Mean net edge	Status
RSI	1 / 5	-0.001150	0.0000678	NOT_ESTABLISHED
Bollinger Bands	1 / 5	-0.001104	0.0000438	NOT_ESTABLISHED
Combined	2 / 5	-0.002617	-0.0001223	NOT_ESTABLISHED

Institutional interpretation

The Version 1 result does not state that all indicator events were incorrect. It establishes that the declared indicator information did not produce stable benchmark-relative forecasting value under the frozen sample, target, cost, and validation design. No deterioration or suspension claim is permitted where prior establishment has not been demonstrated.

Governance and methodological development

The framework distinguishes historical establishment, current validity, structural deterioration, and operational status.

Operational status taxonomy

Status	Institutional definition
NOT_ESTABLISHED	Stable historical incremental value was not demonstrated under the declared validation contract.
ACTIVE	Historical value is established and remains positive under current evidence and monitoring rules.
REDUCED	Historical value is established, but current evidence is uncertain, regime-concentrated, or deteriorating.
SUSPENDED	Historical value is established and both structural-deterioration and recent-performance gates support suspension.

Approved development programme

Phase	Primary objective
II	Conditional validity across horizons, targets, regimes, nonlinear responses, and continuation-versus-reversal interpretations.
III	Time-varying signal effects, fully estimated latent states, duration dependence, and dynamic model averaging.
IV	Bayesian online changepoints, failure probabilities, expected review horizons, and calibrated escalation rules.
V	Funding, open interest, liquidations, order-book information, cross-market transmission, and production governance.

Sources, methodology, and reproducibility

Data. Public Binance spot OHLCV accessed through the CCXT unified interface. The tracked snapshot, validation record, processed features, fold assignments, and SHA-256 checksums support exact replication.

Methodology. RSI follows Wilder; Bollinger variables include %B and BandWidth; market-state filtering is motivated by Hamilton; sequential deterioration follows Page; chronological validation and predictive-comparison controls are documented in the repository references.

Repository. github.com/rolffcoelho-bravo/when-signals-stop-working

Citation. See CITATION.cff and docs/REFERENCES.md. Findings are specific to the declared venue, instruments, sample, target, benchmark, and cost assumption.